Shortest path

PERT

Michel Bierlaire

Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Program Evaluation and Review Technique

We consider an application that can be modeled as a longest path problem, with a network without any cycle.
In that case, the problem can be solved using the shortest path algorithm on the network where the signs of the costs are changed.

PERT

What is the minimum duration of the project?
What are the tasks that do not tolerate any delay without delaying the project?

PERT

Program Evaluation and Review Technique

- Project composed m tasks.
- Each task has a duration.
- Each task has a list of predecessors.

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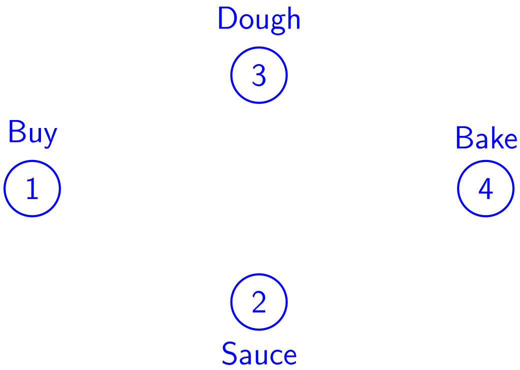
Example: prepare a pizza

1. Buy the ingredients (30 minutes).
2. Prepare the sauce (20 minutes) [1].
3. Prepare the dough (4 hours) [1].
4. Bake (12 minutes) [2,3].

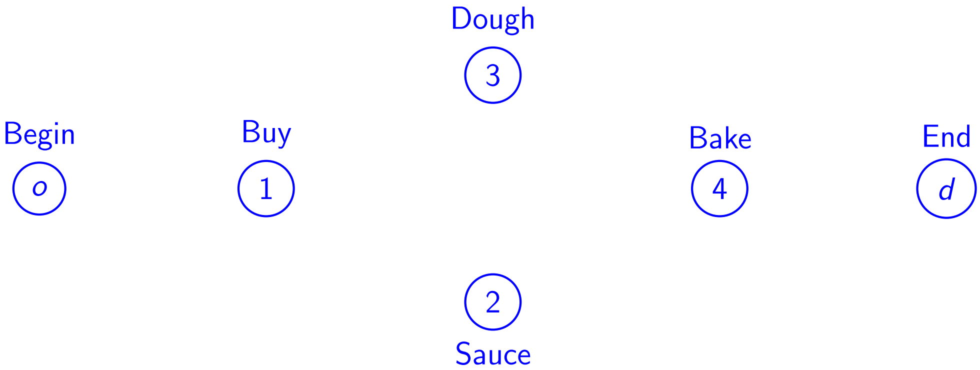
Network

One node per task



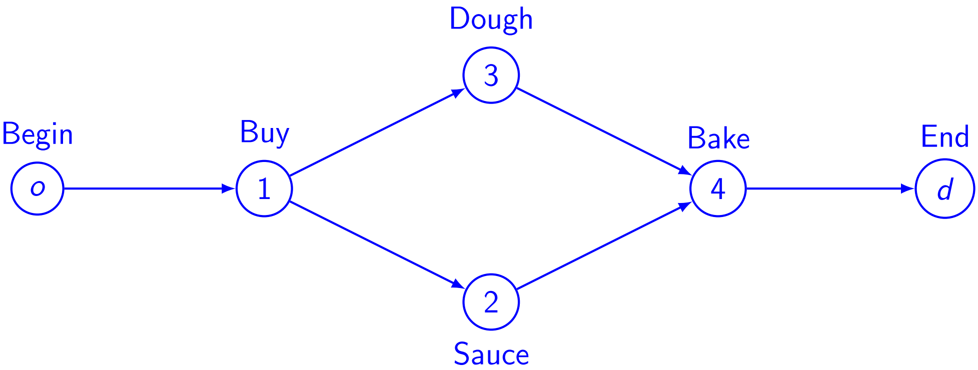
Network

Begin and end



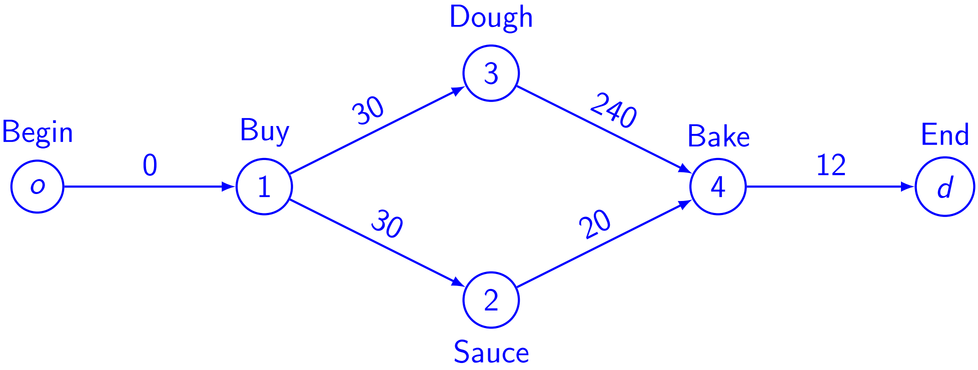
Network

One arc per precedence



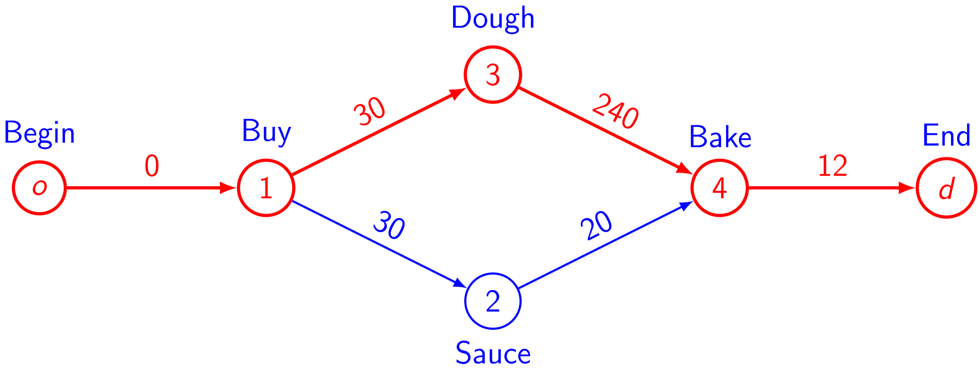
Network

Cost on arc: task duration



Longest path

- What is the minimum duration of the project?
- What are the tasks that do not tolerate any delay without delaying the project?



Summary

In project management, a cycle free network represents the precedence of the tasks.
The longest path in the network identifies the critical tasks, that cannot suffer any delay.
As there is no cycle, a shortest path algorithm can be used to solve the problem.